

ENZYMES

Chapter

2

Major Concept

In this Unit you will learn:

- Enzyme Structure
- Mechanism of Enzyme Action
- Factor effecting Enzyme Action
- Enzyme Inhibition
- Enzyme Classification





Introduction:

Life would not be possible without metabolic activities of the cell. This in turn is dependent upon the catalytic molecules called the enzymes. Without enzymes, the dynamic, steady state of the cell would cease to exist.

Life is a mesh work involving a perfect co-ordination of a vast majority of chemical relations. Some of these reactions result in synthesizing large molecule others in cleaving large molecules and still others either utilize energy or liberate energy. All these reactions would occur very slowly at low temperature and atmospheric pressure. The conditions under which living cells carry on their life processes. In the living system these reactions proceed at extremely high rates. This is due to the presence of some specialized substances or biocatalysts which are synthesized inside the living cells. The biocatalysts are called enzymes (Gr: En; inside, zyme=yeast) the term enzyme was coined by Friedrich Wilhelm Kuhn (1878). Enzymes may be defined as organic substances capable of catalyzing specific chemical reactions in the living system.

Just few years ago, it was considered that all enzyme were proteins. During the 1980s Thomas Cech and Sidney Altman discovered that certain molecules of ribonucleic acid also function as enzymes. These molecules are called Ribozymes which catalyze reactions involved in processing genetic information to be used by a cell but generally enzymes are proteinaceous in nature. Following are some characters of enzymes.

- Enzymes are biocatalyst produced in the protoplasm, synthesized in the cell.
- Most of the enzyme are proteinaceous in nature, macromolecules of globular proteins with high molecular weight.
- They are either consist of protein e.g. amylase or pepsin or may contains, along with protein, a non-protein part e.g. Acetyl CoA.
- Enzyme act with in the cell where they have produced called endo-enzyme and the enzyme which acts outside the cell called exo-enzymes.
- They are specific in nature and function.
- They are much greater is size than the substrate.
- They have particular sites to react with substrate, these sites are called **Active site**.
- Enzyme activity can be accelerated by certain ions or salts called activators e.g. Mn, Mg, Cl^- etc.
- Enzyme can be inhibited by certain factor called inhibitors. e.g. substrate concentration, enzyme concentration, pH and temperature.
- They are sensitive i.e. they work at specific pH.
- They remain chemically unchanged during and after the chemical reactions.

2.1 STRUCTURE OF ENZYME:

As we have discussed that the enzyme are three dimensional globular protein. Every enzyme specifically reacts with special substance called **substrate**.

Active Site:

Each enzyme has a special charged place to fit the substrate called Active Site which is complementary is shape to the shape of substrate. The charge and shape of the active site is formed by some amino acids present in the polypeptide chain of the active site of the enzyme. The amino acids which from active site come closed to arrange in specific way to coil and fold within a globular symmetry.

The active site of enzyme has two regions

- (i) Binding site.
- (ii) Catalytic site.

Binding site is the region of active site whose amino acids make temporary bonds with substrate. On the other hand some amino acids are responsible for catalytic activity therefore it is called catalytic site.

Structural Part:

Beside active site some enzymes are completely made of protein only while the other enzymes are made up of protein and non-protein part. These enzymes are called conjugated enzyme. The non protein part of enzyme is called **co-factor**, it is responsible for attachment of substrate at active site and perform catalytic activity. The co-factors may be organic or inorganic molecules. Inorganic co-factors may be metallic ions. These are detachable co-factors called activators because they get attached at the time of substrate binding. Other co-factors are organic molecules which may divide into two groups i.e. co-enzymes and prosthetic group. **Co-enzymes** are the derivative of vitamin like Nicotine or Flavin in NAD and FAD respectively. They behave like inorganic co-factor i.e. attach at the time of substrate binding and detachable.

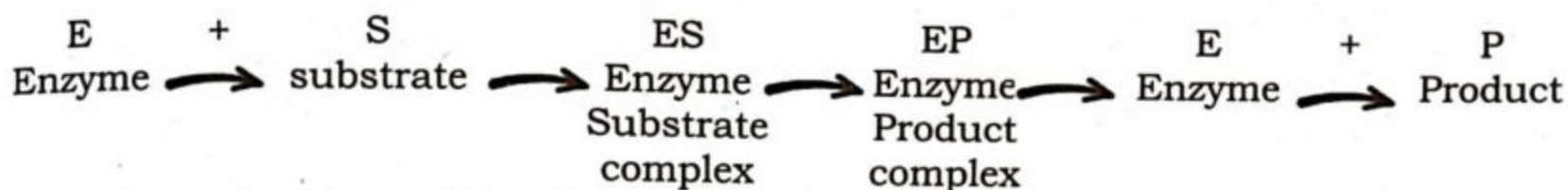
The others type of organic co-factor is **prosthetic group**. It is bonded covalently and permanent part of enzyme i.e. do not detached e.g. Mg containing porphyrin of chlorophyll.

The conjugated enzyme without co-factor called **Apo-enzyme**. It does not work separately. In activated enzyme consists of polypeptide chain and co-factor is collectively called **Holoenzymes**.

Pepsin is an enzyme secreted by gastric gland of stomach in inactive form pepsinogen. Pepsinogen is inactive due to presence of polypeptide chain at its active site to block. When pepsinogen acts with HCl the additional polypeptide removed and become active form pepsin.

2.2 MECHANISM OF ENZYME ACTION:

As we have discussed that all metabolic reactions are catalyzed by enzymes. During these reactions the substrate bind to active site of the enzyme and form enzyme substrate complex (ES) that substrate convert into product but remain attached with enzyme and form enzyme-product complex (EP) and then product release from active site of enzyme. The quantity and shape of enzyme remain same before and after chemical reaction.



In order to explain the mode of action of enzyme two models were presented.

(i) Lock and key model

Emil Fischer 1898 proposed this model, according to this model a particular enzyme acts on a particular substrate like particular lock can be unlocked by a particular key. This theory depend upon physical contact between substrate and enzyme molecule.

The active site of each enzyme has a distinct shape (Rigid) and distribution of charge which is complementary to its substrate like lock and key, where a lock allows very few keys to fit in. Similarly enzymes allow a few complementary molecules to fit in and react while rejecting even fairly similar molecules. These enzymes work in absolute specefity i.e. catalyze only one substrate.

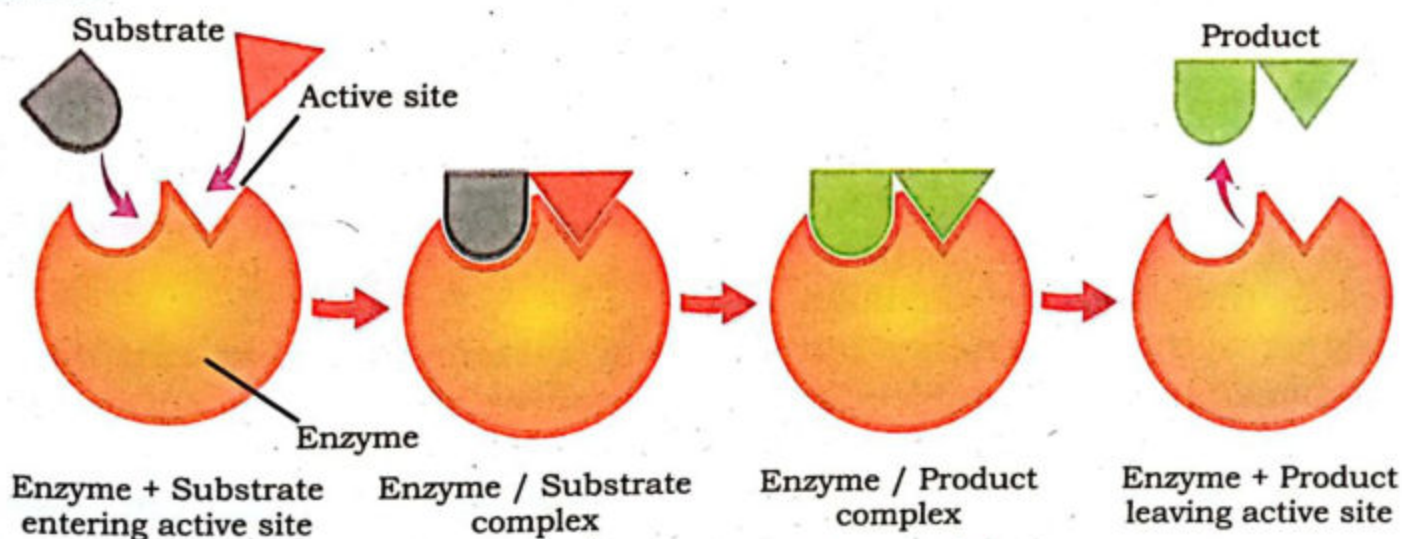


Fig: 2.1 Lock and key model

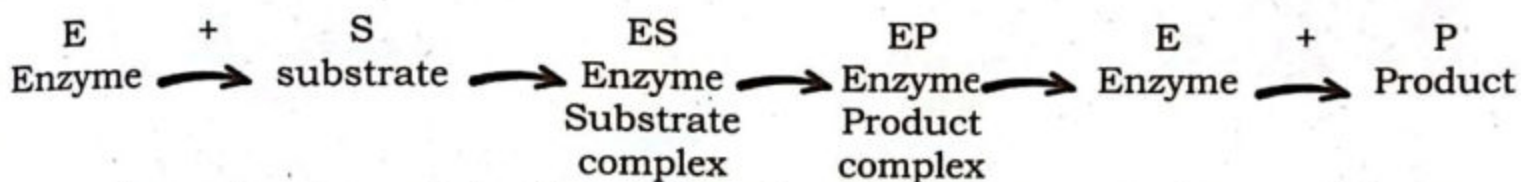
(ii) Induced fit model

Koshland in 1959 proposed another model with the name of induced fit model. He stated that when a substrate combine with an enzyme. It induces change in the enzyme structure, this change enables the enzyme to perform the catalytic activity more efficiently. It means according to his model



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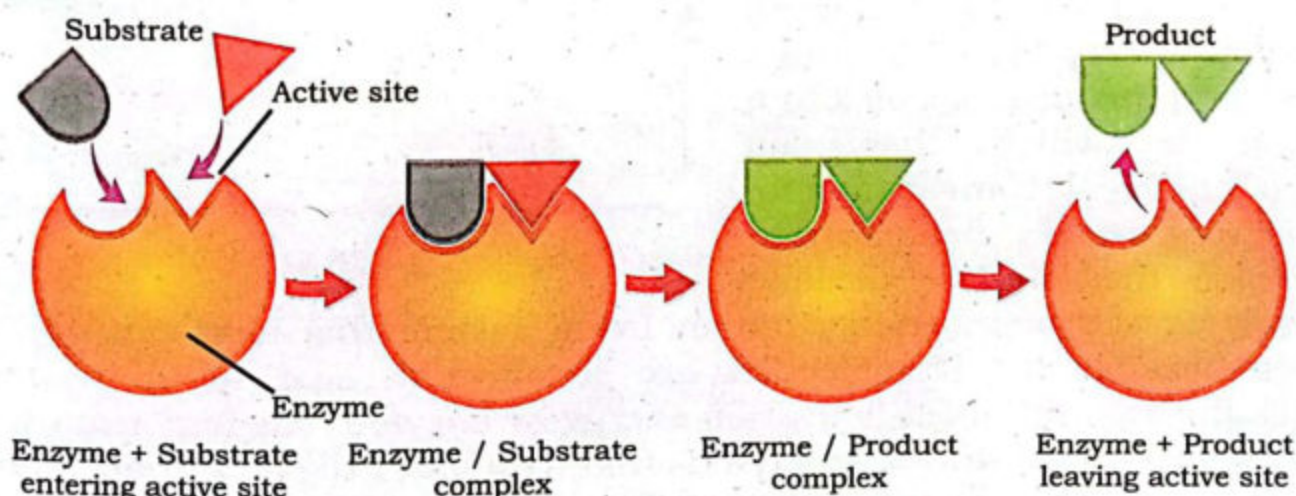


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the active site is flexible. Therefore, the change in active site occur during catalytic reaction is responsible for the conversion of substrate into product.

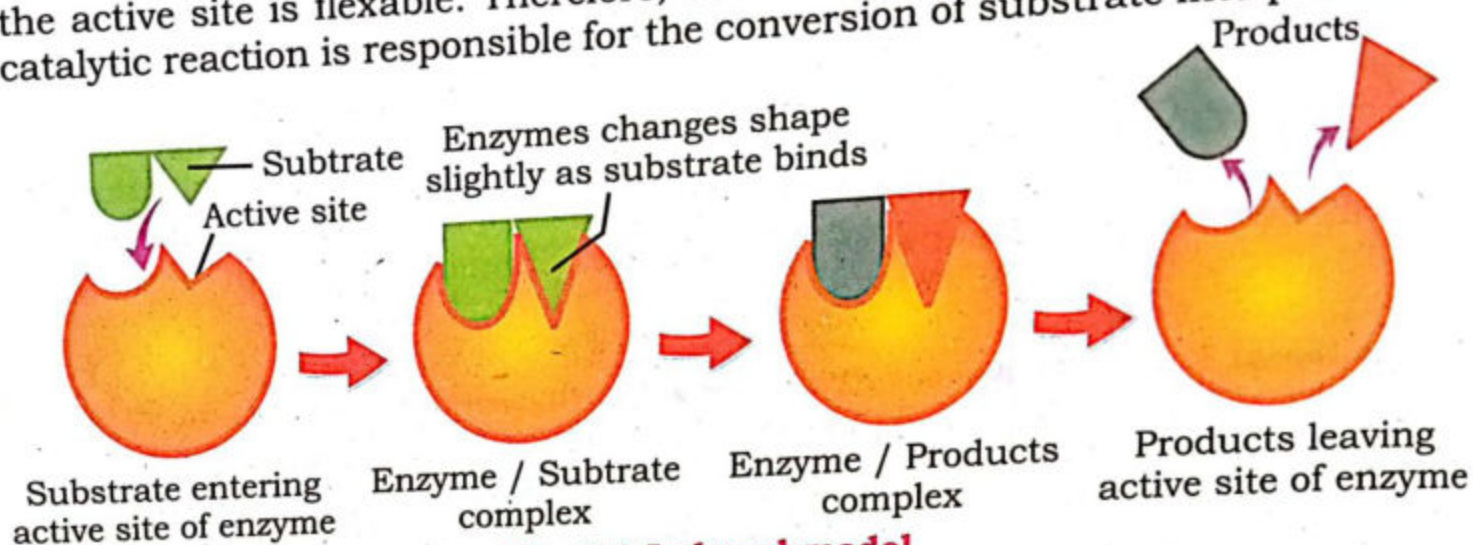


Fig: 2.2 Induced model

2.2.1 Energy of Activation.

The question arises here, how enzyme are able to accomplish such effective catalysis and why thermodynamically favourable reaction do not proceed on their own at relatively rapid rates in the absence of enzyme?

Chemical transformation requires the breakdown of certain covalent bond of substrate. For this, reactants must contain sufficient energy to overcome a barrier, this barrier is energy of activation or **activation energy**. It can be defined as the minimum quantity of energy that is required to activate atoms or molecules to a condition in which they can undergo chemical transformation. In non-living system, heat is used to increase the effective collision and work as activation energy but in living system this heat can not be provided because the biomolecules are sensitive to heat so the system decrease the amount of his activation energy by enzyme. The important role played by the enzyme during reaction is that they lower the activation energy of the reaction. The enzyme reacts with the energy rich and energy poor molecules to form an intermediate complex. This complex again breaks into product and enzyme. If activation of this complex is low many molecules can participate in reaction. In this way activation energy is lowered by the enzyme but in this action equilibrium i.e. ratio of reactant and product concentration is never altered.

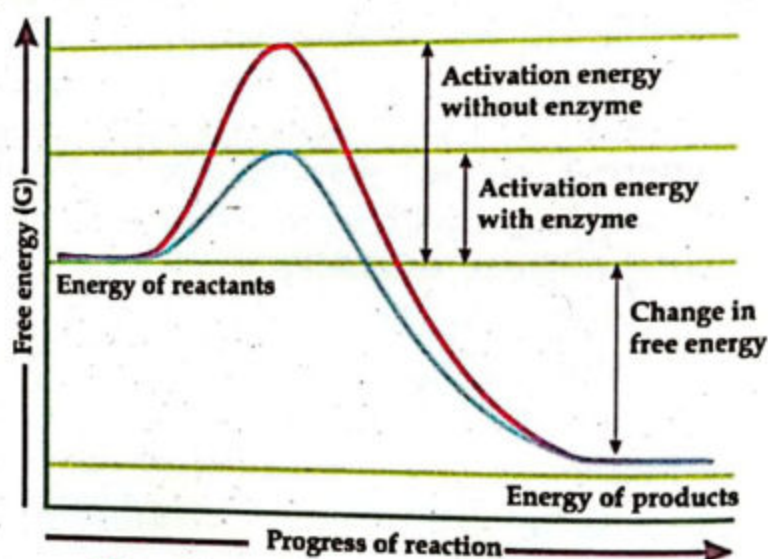


Fig: 2.3 Energy of activation



2.3 FACTORS AFFECTING ENZYME ACTIVITY

The rate of enzyme activity means the quantity of substrate transform into product in a unit time (1 second). Greater the amount of product formed per second greater will be the enzyme activity or vice versa. Some of the factors which affect the rate of enzyme activity are temperature, pH concentration of enzyme and concentration of substrate etc.

When an inhibitor bind to an enzymes away from active site the shape of enzyme become alter due to these allosteric inhibitor. As a result of this change all active sites of the enzymes are changed slightly so that they become unable to receive substrate. Some molecules also work as activator when bind to the enzyme away from the active site these molecules are called allosteric activators, these allosteric activators an increase in the function of active site. These molecules do not bound covalently to the enzymes so their interactions are reversible. These activators may influenced by thermal factor and concentration of substrates.

2.3.1 Temperature

The chemical reactions depend on the molecular motion, i.e. higher the number of substrate molecules reach to enzyme higher will be the chances of reaction or vice versa. It is estimated that increase in 10°C of temperature double this rate of reaction. Enzymes are also sensitive to temperature due to its proteinecous composition. The temperature where enzyme work maximumly or rate of enzyme activity is maximum called optimum temperature, below and above this temperature the enzyme activity decreases. The human enzyme are highly active at about 37°C and all are completely denatured at 100°C , where at minimums i.e. 0°C , activity is reduced to minimum but enzymes are not destroyed. Most of the enzymes in higher organism have optimum temperature between 25°C to 42°C .

Some bacteria which survive in hot spring are called thermophilic because they have optimum temperature of 70°C or higher. Such enzymes are also used in biological washing powder at high temperature washing.

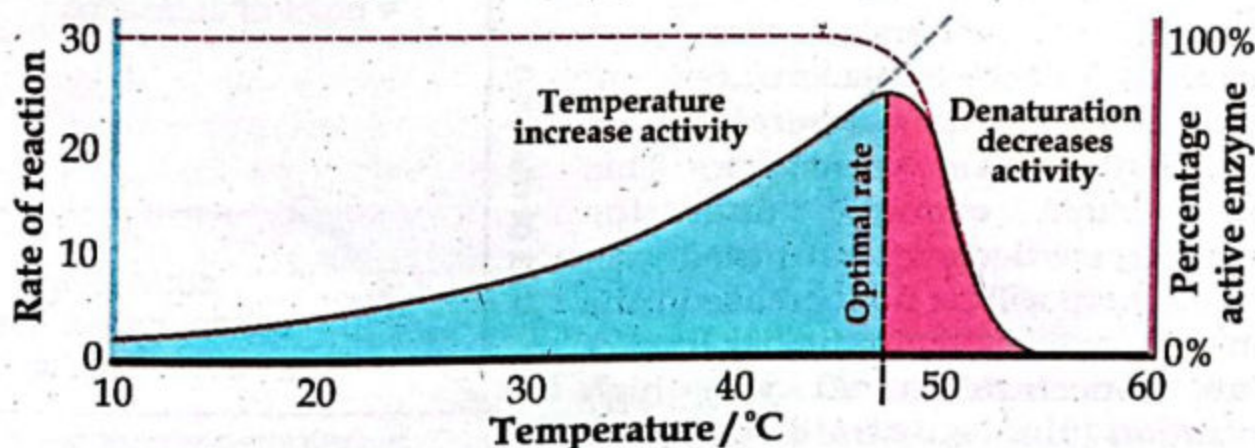


Fig: 2.4 Effect of temperature on enzymes

2.3.2 Effect of pH:

The activity of enzyme varies considerably with pH and there is generally a marked optimum pH for each enzyme. Each enzyme has very narrow range or it works as optimum pH. The pH has marked effect on the charges of an amino acid side chains which form tertiary and quaternary structure of protein (enzyme). Little change in pH causes ionization of amino acids of enzyme and they becomes inactive.

Majority of the human digestive enzyme works in the range of 7 to 8. With few exception like gastric enzyme secreted by some cells of stomach i.e. pepsin. The pepsin works at highly acidic condition i.e. its optimum pH is 1.4 but works in the range of 1.5 to 2.5. On the other side amylase is the enzyme released from pancreas work at highly alkaline i.e. 8.5 pH.

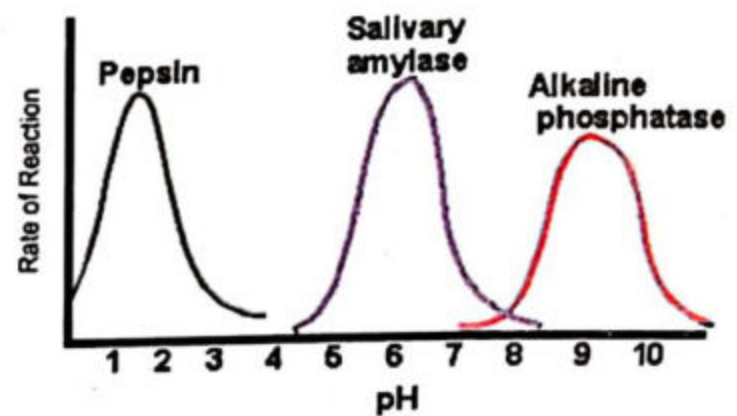


Fig: 2.5 Effect of pH

2.3.3 Enzyme Concentration:

The enzyme activity depends upon the availability of substrate and enzyme molecules. If the substrate concentration is very high the enzyme concentration is directly proportional to the concentration of enzyme but in real condition the concentration of substrate is always limited so the enzyme activity increase with the increase in the enzyme concentration initially but after some time it remains constant.

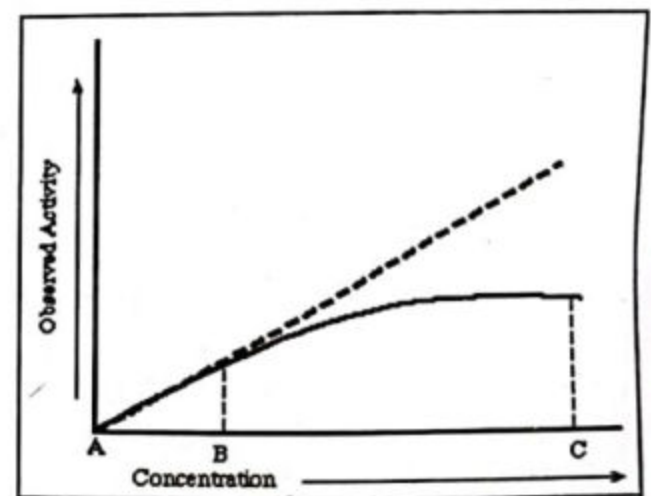


Fig: 2.6 Enzyme concentration graph

2.3.4 Substrate Concentration:

The rate of reaction increases with an increase in the concentration of substrate in a condition where enzyme concentration remain constant. This increase remain continue until the available enzyme becomes saturated with substrate. There will be no increase in the enzyme activity to a certain higher level of substrate concentration. At very high concentration the substrate exerts a retarding effects upon enzyme action.

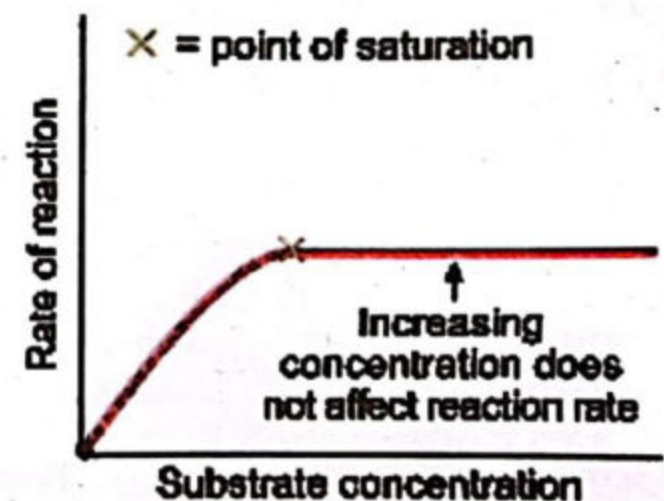


Fig: 2.7 Substrate concentration graph



This may be due to two reasons:

- (a) Higher quantity of substrate than enzyme.
- (b) Accumulation of end product.

It means the substrate and enzyme concentration are directly proportional up to a certain maximum velocity after which further increase in substrate concentration has no effect on the rate of reaction.

2.4 ENZYME INHIBITION:

Sometimes enzyme does not work properly or system wants to decrease this enzyme activity because the products are no more required this condition of decrease in enzymatic related processes, enzyme production or enzyme activity is called **enzyme inhibition**.

2.4.1 Types of Enzymes Inhibition:

There are three types of enzyme inhibition

- i) Competitive inhibition ii) Non-competitive inhibition
- iii) Un competitive inhibition

Substances which decrease the activity of an enzyme are called inhibitors. The inhibitor may act by combining directly with the enzymes or they may react with the activators therefore, activator does not remain available to enzyme for activation.

(i) Competitive Inhibition:

The type of inhibition occurs by the molecules which are very similar to the substrate and bind to active site instead of original substrate. These molecules block active site for actual substrate e.g. Penicillin is a competitive inhibitor which block the enzyme which is responsible to construct bacterial cell-wall. If this, inhibition is reversible it can overcome by increasing concentration of substrate so that as active site become available more substrate molecule than inhibitor molecules are around to gain entry to these sites.

(ii) Non-Competitive Inhibition:

The non-competitive inhibition is the type of inhibition when a molecule binds with enzyme at any site except the active site. These molecules are called non-competitive inhibitors. These binding sites which are away from active sites are called **allosteric site** (Allos=away). This interaction causes the enzyme molecule to change its shape, reading the active site unresponsive to the substrate or leaving the enzyme less effective at catalyzing for the conversion of substrate to product. Feedback inhibition is the example of reversible non-competitive enzyme inhibition.

(iii) Un-Competitive Inhibition:

It is also known as anticompetitive inhibition. It takes place when an enzyme's inhibitor binds only to the complex formed between the enzyme and the substrate (E-S-Complex) this type of inhibition typically occurs in reactions with two or more substrate or product.

2.4.2 Significance of Enzyme Inhibition:

It serves as a major control mechanism of biological systems especially when inhibition occurs by small molecules. It is also used as strategy for drug discovery. It also gives knowledge into the metabolic path way e.g. identify substrate and condition critical for catalysis. Enzyme inhibitors are used to screen various levels of diseases which propel the growth of inhibitors.

2.4.3 Feed Back Inhibition:

The activity of almost every enzyme in a cell is regulated by feedback inhibition. It is a type of biological control mechanism where the activity of an enzyme is inhibited by its own product. It is called feedback inhibition. It is reversible non-competitive inhibition and takes place during metabolic path way. It usually occurs, when the product is in high quantity, it binds competitively with its enzymes active site or come in between. Substrate and enzyme once the product is consumed the inhibition also reduced and form more product.

Most enzymatic path ways are also regulated by feedback inhibition, but in these cases the end product of path way binds at an allosteric site on the first enzyme of the path way. This binding shutdown the path way and no more product is produced.

2.5 CLASSIFICATION OF ENZYMES:

Enzyme can be classified on different basis either on the basis of the types of reactions they performed or on the basis of substrate at which they act.

Classification on the basis of types of reaction:

They can be classified in following group

1) Oxido Reductase:

Enzymes require to perform oxidation and reduction reaction by removal and addition of H^+ respectively e.g. Ferredoxin reducing substance.

2) Transferase:

Enzymes responsible to transfer a group from one substance to other i.e. transfer of phosphate from ATP to hexose.

3) Hydrolase:

These are digestive enzyme responsible to break the bond by H_2O and convert oligomers into monomers.

4) Lyase:

These enzymes catalyze the breakdown of specific covalent bonds and removal of group without hydrolysis.

5) Isomerase:

Enzyme catalyse intra molecular re-arrangement of atoms in the molecules to form another isomer e.g. glucose - 6 - phosphate converted into fructose -6-phosphate during glycolysis by enzyme phosphohexose.



6) Ligase (Synthetase)

Enzyme Catalyse the condensation reaction like DNA and RNA polymerase.

2.5.1 Classification based upon substrate:

- i) Protease – act on protein e.g. trypsin, pepsin e.t.c.
- ii) Lipase – perform hydrolysis of fats into glycerol and fatty acids.
- iii) Glycosidase – act on carbohydrate molecules like
Amylase – act on amylose to convert into maltose.
Cellulase – act on cellulose
Maltase – act on maltose.
Sucrase – act on sucrose.
Lactase – act on lactose.
- iv) Nuclease – act on different nucleic acids.
e.g. RNAase, DNAase, ATPase etc.

SUMMARY

- Organic substance capable of catalyzing specific chemical reactions in the living system called enzyme.
- Most of the enzymes are made up of protein and very few are of RNA. They are globular in nature.
- The enzyme which act within a cell called endo enzyme and those which act outside the cell called exo enzyme.
- Enzyme are specific in nature and function.
- Each enzyme has a particular site where substrate attach with it called active site.
- The active site has two sites (a) Binding site (b) Catalytic site.
- The enzyme which are made up of protein and non-protein part called conjugated enzymes. This non-protein part is called co-factor.
- For enzyme action two models are presented.
- (a) Key lock model (b) Induced fit model.
- Activation Energy is the minimum amount of energy that is require to activate atoms or molecule for a reaction.
- Enzyme activity is affected by temperature, pH, Enzyme, concentration, substrate concentration etc.
- Decrease in enzyme activity or inactivation of enzyme is called enzyme inhibition.
- Three types of enzyme inhibition are there (i) competitive (ii) non-competitive (iii) uncompetitive inhabitation.
- The type of biological control where enzyme activity is inhibited by its own product called feedback inhibition.

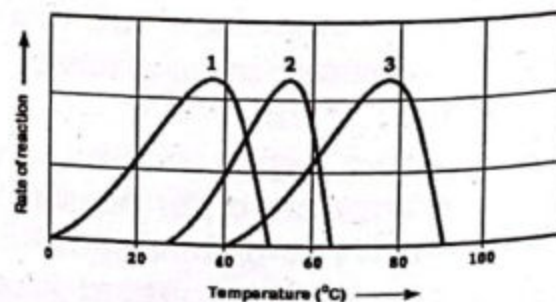


- Oxido-reductase, Transferase, Hydrolase, Lysase, Isomerase, Ligase are the type of enzyme on the basis of reactions.
- Protease, Lipase, Glycosidase, Nuclease are the types of enzyme on the basis of substrate on which enzyme act.

EXERCISE

1. Encircle the correct choice

- (i) Which of the following statements is (are) true about enzyme catalyzed reactions?
 - (a) The reaction is faster than the same reaction in the absence of the enzyme.
 - (b) The free energy change of the reaction is opposite from the reaction in the absence of the enzyme.
 - (c) The reaction always goes in the direction toward chemical equilibrium.
 - (d) Both a and b.
- (ii) The active site of an enzyme is the region that
 - (a) Binds allosteric regulators of the enzyme
 - (b) Is involved in the catalytic reaction of the enzyme
 - (c) Binds the products of the catalytic reaction
 - (d) Is inhibited by the presence of a coenzyme or a cofactor.
- (iii) According to the induced fit hypothesis of enzyme catalysis, which of the following is correct?
 - (a) The binding of the substrate depends on the shape of the active site.
 - (b) Some enzymes change their structure when activators bind to the enzyme.
 - (c) A competitive inhibitor can outcompete the substrate for the active site.
 - (d) The binding of the substrate changes the shape of the enzyme's active site.
- (iv) Which curve represents the behavior of an enzyme taken from a bacterium that lives in hot-springs.
 I. Curve 1 II. Curve 2 III. Curve 3
 - (a) I only
 - (b) II only
 - (c) III only
 - (d) I and II
 - (e) II and III





- (v) Increasing the substrate concentration in an enzymatic reaction could overcome which of the following?
- (a) Denaturization of the enzyme
 - (b) Allosteric inhibition
 - (c) Competitive inhibition
 - (d) Saturation of the enzyme activity
- (vi) Competitive inhibitors block the entry of substrate into the active site of an enzyme. On which of the following properties of an active site does this primarily depend?
- (a) The ability of an enzyme to form a template for holding and joining molecules
 - (b) The enzyme's ability to stretch reactants and move them toward a transition state
 - (c) The enzyme providing an appropriate microenvironment conducive to a reaction's occurrence
 - (d) The enzyme forming covalent bonds with the reactants
- (vii) A series of enzymes catalyze the reaction $X \rightarrow Y \rightarrow Z \rightarrow A$. Product A binds to the enzyme that converts X to Y at a position remote from its active site. This binding decreases the activity of the enzyme. What is substance X?
- (a) A coenzyme
 - (b) An allosteric inhibitor
 - (c) A substrate
 - (d) An intermediate
- (viii) If an enzyme is added to a solution where its substrate and product are in equilibrium, what would occur?
- (a) Additional product would be formed.
 - (b) Additional substrate would be formed.
 - (c) The free energy of the system would change.
 - (d) Nothing; the reaction would stay at equilibrium.
- (ix) Which one of the following is the mechanism of action of enzymes?
- (a) They act upon substrate molecules to release new substrate molecules.
 - (b) They actually increase the amount of energy of activation.
 - (c) Enzymes dramatically decrease the amount of energy of activation.
 - (d) Enzymes break product molecules to release new product molecules.



- (x) How does a noncompetitive inhibitor decrease the rate of an enzyme reaction?
- By binding at the active site of the enzyme
 - By changing the shape of a reactant
 - By changing the free energy change of the reaction
 - By acting as a coenzyme for the reaction

2. Write short answers of the following questions:

- Why enzymes are specific in nature?
- Why enzyme activity is directly proportional to enzyme concentration?
- Why enzymes are called temperature sensitive?
- How enzyme reduces the energy of activation?
- How enzymes activity effect on substrate concentration?
- What do you mean by prosthetic group?
- Differentiate between the activator and inhibitors.

3. Write detailed answers of the following questions:

- What is enzyme? explain characteristics of enzyme.
- Describe the effect if temperature on the rate of enzyme action.
- Describe mechanism of enzyme action.
- Describe the classification of enzymes.
- Describe enzymatic inhibition, its types and its significance.
- Describe structure of enzyme.
- Explain the effect of substrate and concentration on the rate of enzyme action.

CELL STRUCTURE AND FUNCTION

Chapter

3

Major Concept

In this Unit you will learn:

- Techniques used in Cell-Biology
- Cell-Wall and Plasma membrane as boundary wall
- Cytoplasm and Organelles
- Prokaryotic and Eukaryotic Cells

